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DEPARTMENT OF COMPUTER SCIENCE ASSIGNMENT - I

NAME	:	M.ABDUL MAZEETH
REGISTER NUMBER	:	124012173061
CLASS	:	B.Sc., ARTIFICIAL INTELLIGENCE
SEMESTER	:	V
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Terminology of Neural Network and Types of Activation Functions

Artificial Neural Networks (ANNs)

Introduction

Artificial Neural Networks (ANNs) are computational models inspired by the structure and functioning of the human brain. They are widely used in Machine Learning and Artificial Intelligence for tasks such

as image recognition, speech processing, natural language processing, and prediction systems. A neural network consists of interconnected nodes called neurons that work together to process information and learn patterns from data.

Terminology of Neural Network

Understanding the basic terminology of neural networks is essential for learning deep learning concepts.

1. Neuron (Node)

A neuron is the fundamental processing unit of a neural network. It receives inputs, processes them, and produces an output. Similar to biological neurons, artificial neurons communicate with each other through connections.

- Receives input data
- Multiplies inputs by weights
- Adds bias
- Applies an activation function
- Produces output

2. Input Layer

The input layer is the first layer of a neural network. It receives raw data from the outside world and passes it to the hidden layers.

- Image pixels
- Text data
- Sensor readings

3. Hidden Layer

Hidden layers are located between the input and output layers. These layers perform computations and extract important features from the data.

- Learn complex patterns
- Improve prediction accuracy
- Enable deep learning models

4. Output Layer

The output layer provides the final result of the neural network.

- Classification of images
- Prediction of prices
- Sentiment analysis results

5. Weights

Weights determine the importance of each input. During training, the network adjusts weights to minimize errors.

- Higher weight → greater influence
- Lower weight → lesser influence

6. Bias

Bias is an additional parameter added to neurons. It helps the network fit the data more effectively and improves learning capability.

- Increases flexibility
- Improves accuracy
- Helps in shifting activation functions

7. Activation Function

An activation function determines whether a neuron should be activated or not. It introduces non-linearity into the network.

- Enables learning of complex patterns
- Helps make decisions
- Improves model performance

8. Epoch

An epoch represents one complete cycle through the entire training dataset.

- Example: If a dataset contains 1000 samples and the network processes all samples once, it completes one epoch.

9. Learning Rate

The learning rate controls how much the weights are updated during training.

- High learning rate: Faster learning but may miss optimal solutions.
- Low learning rate: Slower learning but more accurate.

10. Loss Function

A loss function measures the difference between actual output and predicted output.

- Mean Squared Error (MSE)
- Cross Entropy Loss
- Hinge Loss

Types of Activation Functions

Activation functions play a crucial role in neural networks. They decide how information flows through the network.

1. Binary Step Function

The Binary Step Function produces either 0 or 1 as output.

Formula:

- Output = 1 if $x \geq 0$
- Output = 0 if $x < 0$

Advantages:

- Simple implementation
- Easy to understand

Disadvantages:

- Cannot handle complex problems
- Not suitable for deep learning

2. Linear Activation Function

The output is directly proportional to the input.

Formula:

- $f(x) = x$

Advantages:

- Simple computation

Disadvantages:

- Cannot solve non-linear problems
- Limited learning capability

3. Sigmoid Function

The Sigmoid function converts values into a range between 0 and 1.

Formula:

- $f(x) = 1 / (1 + e^{-x})$

Advantages:

- Useful for binary classification
- Produces probability values

Disadvantages:

- Vanishing gradient problem
- Slow convergence

4. Tanh Function

The Tanh function outputs values between -1 and 1.

Formula:

- $f(x) = \tanh(x)$

Advantages:

- Zero-centered output
- Better than sigmoid in many cases

Disadvantages:

- Still suffers from vanishing gradients

5. ReLU (Rectified Linear Unit)

ReLU is one of the most widely used activation functions.

Formula:

- $f(x) = \max(0, x)$

Advantages:

- Fast computation
- Reduces vanishing gradient problem
- Widely used in deep learning

Disadvantages:

- Dead neuron problem

6. Leaky ReLU

Leaky ReLU allows small negative values instead of zero.

Formula:

- $f(x) = x$, if $x > 0$
- $f(x) = 0.01x$, if $x < 0$

Advantages:

- Solves dead neuron problem
- Better gradient flow

Disadvantages:

- Slightly more complex

7. Softmax Function

Softmax converts outputs into probability values whose sum equals 1.

Applications:

- Multi-class classification
- Image recognition
- Language processing

Advantages:

- Provides probability distribution
- Useful for classification tasks

Activation Function Illustrations

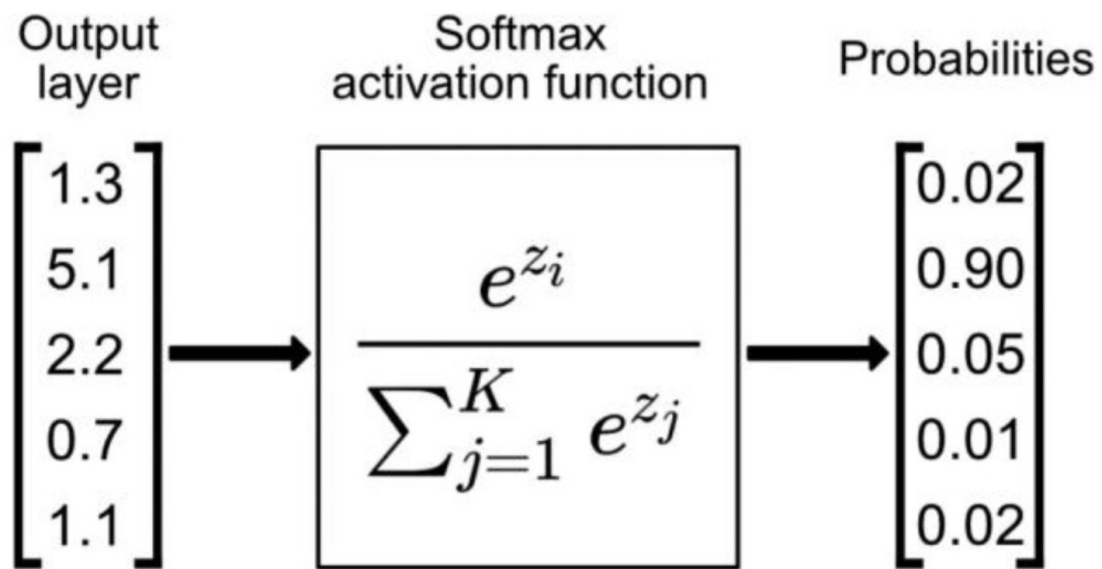


Figure 1: Softmax activation and probability distribution

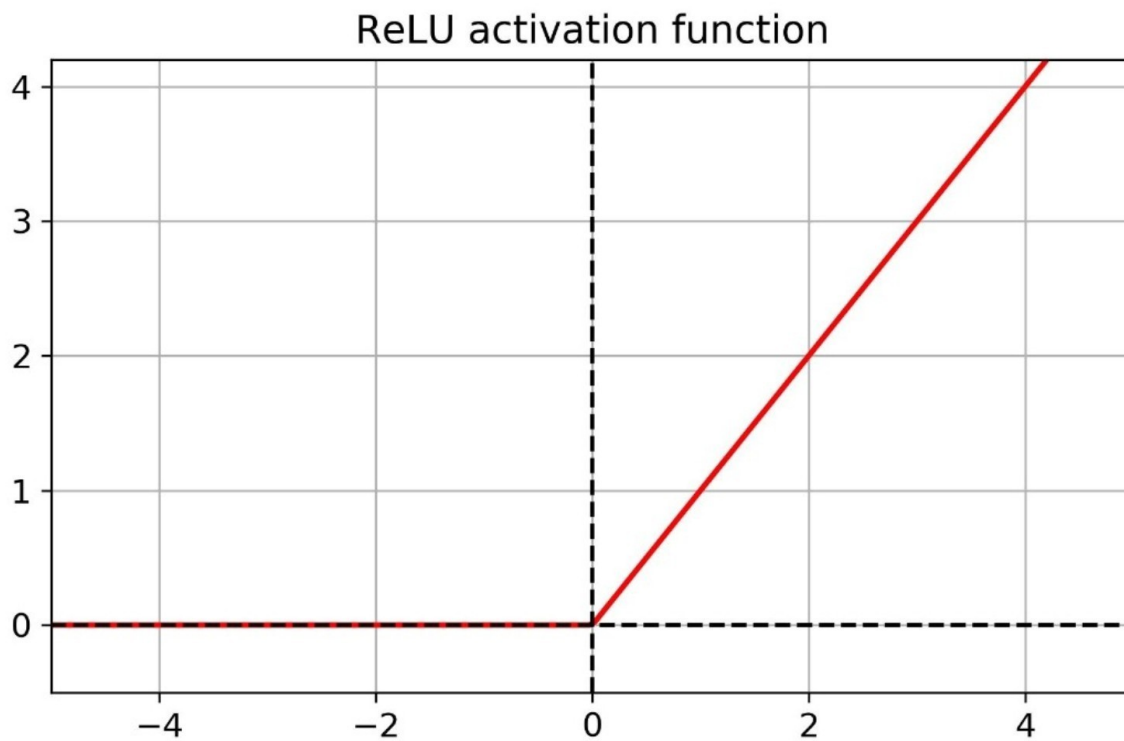


Figure 2: ReLU activation function

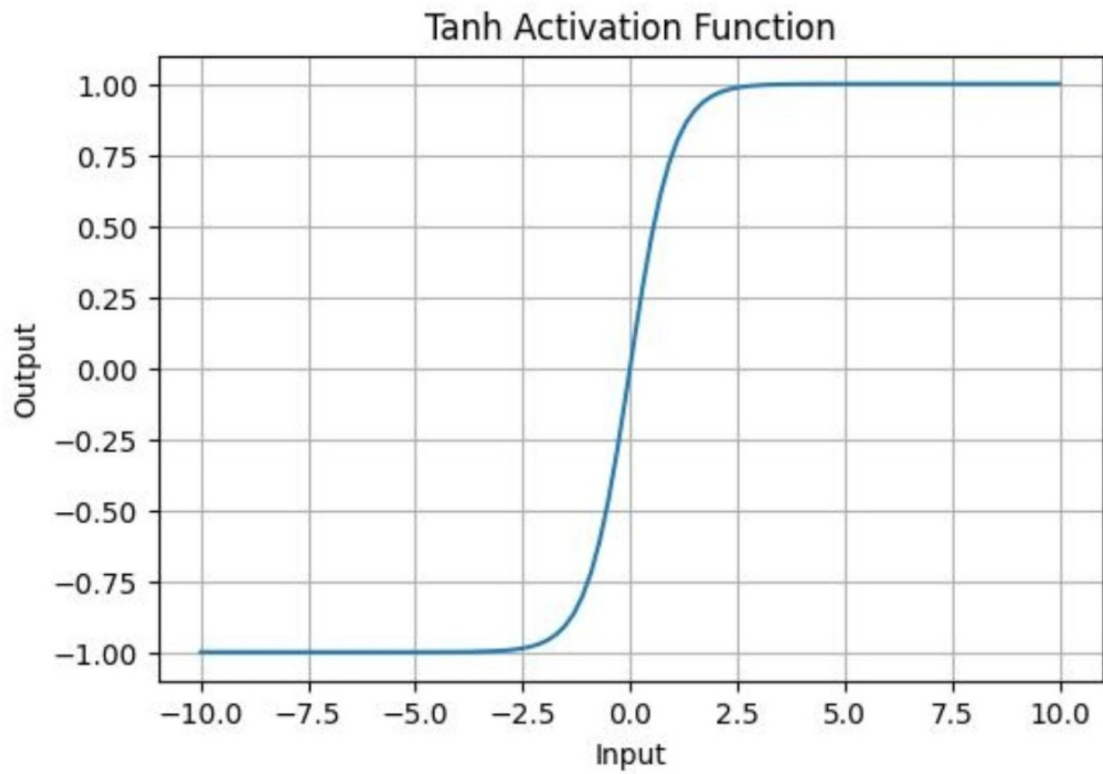


Figure 3: Tanh activation function

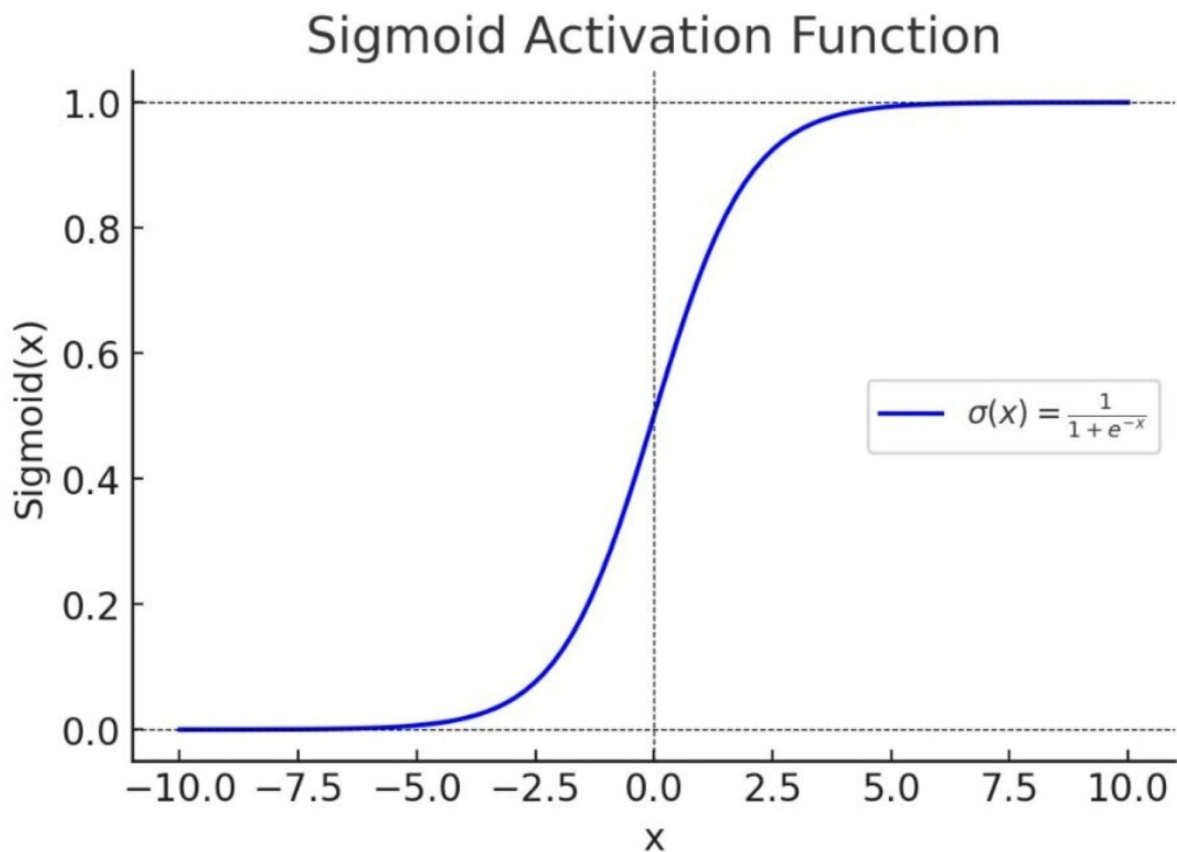


Figure 4: Sigmoid activation function

Applications of Neural Networks

1. Image Recognition
2. Speech Recognition
3. Medical Diagnosis
4. Fraud Detection
5. Self-Driving Cars
6. Recommendation Systems
7. Natural Language Processing
8. Robotics

Advantages of Neural Networks

- Learns complex patterns
- Handles large datasets
- High prediction accuracy
- Supports automation
- Useful in AI applications

Conclusion

Neural Networks are powerful machine learning models inspired by the human brain. They consist of neurons, layers, weights, biases, activation functions, and learning mechanisms that enable intelligent decision-making. Activation functions such as Sigmoid, Tanh, ReLU, Leaky ReLU, and Softmax play a vital role in improving network performance. Due to their ability to learn from data and solve complex problems, neural networks have become a foundation of modern Artificial Intelligence and Deep Learning systems.